

# Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres with a CHIME FRB forecast

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Accepted XXX. Received YYY; in original form ZZZ

## ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids,  $z < 0.114$ , median  $R_{\max} = 11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$ , median  $R_{\text{eff}} = 15.5 h^{-1} \text{Mpc} = 22.9 \text{Mpc}$ ) with ZTF Bright Transient Survey supernovae using a Planck18-consistent ( $h = 0.6766$ ) KD-tree ball query with 1-deg SDSS mask (quantized; edge test in text). Of 6873 BTS transients at  $z = 0.01\text{--}0.114$ , 2465 are retained (2422 mask-classified: 147 void ( $r/R_{\max} < 0.8$ ), 195 shell (0.8–1.0), 2080 wall; 43 outside search radius). Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 142 void (118 SN Ia; subtype audit pending TNS). Volume-limited  $z \leq 0.06$ : 1270 SNe total, 69 void (46 Ia). Raw sensitivity to a void–wall Hubble offset is  $\sim 0.05 \text{ mag}$  at 95 per cent stat-only (matched mass-controlled sensitivity  $\sim 0.07 \text{ mag}$ ); blinded SALT2 standardisation with mass, peculiar-velocity and selection controls is in progress. CHIME Catalog 1 (600 bursts, 524 sources) is used for footprint and error-budget demonstration only; unlocalised  $z < 0.114$  stacking is underpowered by  $> 10\times$  and joint constraints are deferred to Catalog 2. Catalog, randoms design and pipeline will be released.

**Key words:** large-scale structure of Universe – supernovae: general – fast radio bursts – methods: statistical

## 1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Table 1 summarises prior work: void-transient demographics (Wang et al. 2026; Tsaprazi et al. 2021), ZTF Ia environments (Aubert et al. 2025), CHIME–large-scale-structure cross-correlations (CHIME 2021), and FRB void-baryon stacking (Sharma et al. 2026). No published paper jointly uses VoidFinder DR7 spheres (Douglass et al. 2023) with flux-limited BTS (Perley et al. 2020; Fremling et al. 2019) and CHIME in the same low-redshift volumes; this work provides a supernova-main result with an FRB forecast in that geometry. Distances assume Planck Collab. (2020) Planck18 ( $H_0 = 67.66 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_m = 0.3097$ ); void-catalog lengths in  $h^{-1}$  units use  $h = 0.6766$  throughout.

## 2 DATA

VAST Zenodo 7406035 v1.3.0 (Douglass et al. 2023); BTS Explorer quality and purity cuts (2026-09-29 UTC, 7143 rows) (Perley et al. 2020; Fremling et al. 2019); ZTF SN Ia DR2 SALT2 parameters and host masses (Rigault et al. 2024) for the Hubble arm; CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources) (CHIME 2021) with Catalog 2 (CHIME 2026) for exposure; NASA-Sloan Atlas v1\_0\_1 downloading for parent masses; Planck Collab. (2020) throughout.

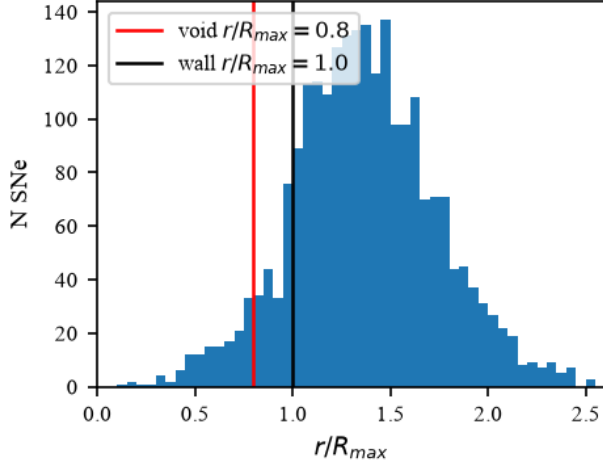
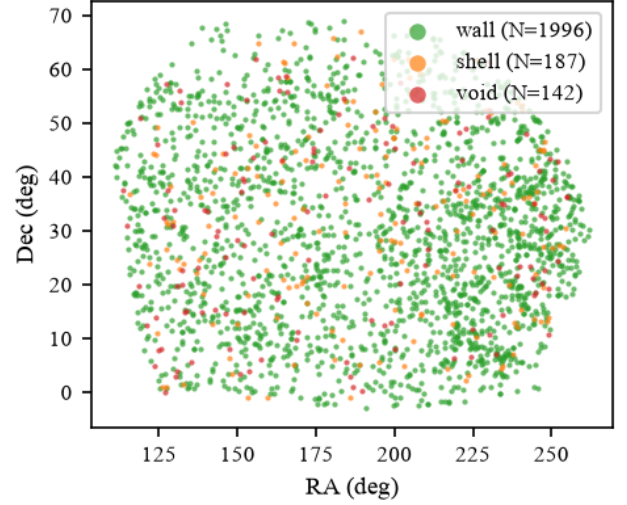
## 3 METHODS

Void centres recomputed from RA, Dec, and comoving distance  $D$  (catalogue  $h^{-1}$  lengths converted with  $h = 0.6766$ , Planck18  $H_0 = 67.66$ ) in one consistent frame; stored-XYZ residual median 0.00 Mpc (p90 0.01 Mpc).  $R_v \equiv R_{\max}$  core spheres (median  $11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$ );  $R_{\text{eff}}$  systematics branch gives void 408/shell 622/wall 1337 (vs 142/187/1996), confirming  $R_{\max}$  is the conservative core choice. Membership via ball query,  $r/R_{\max} = \min$ , shell 0.8–1.0 excluded from primary. Mask from NSA\_main\_mask.pickle (360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts  $0.0600 \rightarrow 0.0615$  (2.5% relative, subdominant to Poisson 9%). Randoms: 200000 uniform comoving ( $z = 0.01\text{--}0.114$ ) per seed (42,43,44),  $\sim 35200$  in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected  $f_{\text{void}} = 0.1896 \pm 0.0018$  (seed spread),  $f_{\text{shell}} = 0.166$ ,  $f_{\text{wall}} = 0.627$  finite ( $f_{\text{out}} = 0.018$ );  $V_{\text{eff,tot}} = 8.80 \times 10^7 \text{ Mpc}^3$ . Observed counts read from the parquet (no hand copies). Blinded SALT2 (Guy et al. 2007) hierarchical  $\Delta_{\text{HR}}$  with Tripp standardisation (Tripp 1998), peculiar-velocity covariance, mass-step joint fit and BTS efficiency planned; this version reports mask-corrected counts with randoms volumes, split apparent rate ratios, and blinded DR2 pilot differences (all blinded values quoted with sealed absolute scale). Estimator:  $R_{\text{env}} = (N_{\text{env}}/V_{\text{eff}} f_{\text{env}})/(N_{\text{wall}}/V_{\text{eff}} f_{\text{wall}})$  with Poisson+Monte Carlo errors; sensitivity  $\text{SE} = \sigma \sqrt{1/N_v + 1/N_w}$  ( $\sigma = 0.15$ ,  $N_v = 118$  gives 0.028 stat,  $\sim 0.035$  with floor; matched  $N = 27$  gives  $\sim 0.07$ ).

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**Table 1.** Prior void-transient work vs this work. Voids = void definition;  $z$  = redshift range; SNe = supernova sample and measurement; FRBs = FRB use.

| Study       | Voids                          | $z$        | SNe                                   | FRBs                    |
|-------------|--------------------------------|------------|---------------------------------------|-------------------------|
| Wang+26     | lumped empty spheres           | $< 0.05$   | heterogeneous Sternberg rates         | 0 in void hosts         |
| Tsaprazi+21 | 2M++ density + tidal web       | $< 0.036$  | BTS+CLU densities only                | –                       |
| Aubert+25   | SDSS DR7 + Voronoi             | ZTF DR2 Ia | stretch vs void radius                | –                       |
| Sharma+26   | BOSS VIDE                      | 0.2–0.7    | –                                     | 3455-sightline DM stack |
| This work   | VAST VoidFinder DR7 $R_{\max}$ | $< 0.114$  | BTS counts + DR2 $\Delta_{\text{HR}}$ | Cat1 error budget       |

**Figure 1.**  $r/R_{\max}$  for  $N = 2367$  cosmology-grade SNe (void  $r/R_{\max} < 0.8$ ,  $N = 142$ ; wall  $> 1.0$ ). Vertical lines mark the analysis thresholds.**Figure 2.** Masked RA–Dec for cosmology-grade SNe: wall ( $N = 1996$ , green), shell ( $N = 187$ , orange), void ( $N = 142$ , red). Outside-mask and edge-excluded objects omitted.

## 4 RESULTS

Cuts flow: 7143  $\rightarrow$  6873 ( $z$ )  $\rightarrow$  2465 retained (2422 mask-classified + 43 outside search radius; 4408 fall outside the SDSS DR7 footprint entirely). Cosmo SNe 2367: void 142, wall 1996 (187 shell, 42 outside; Table 2). Observed void fraction  $142/2367 = 0.060$  versus randoms 0.190: apparent (uncorrected) void/wall ratios 0.24 all, 0.26 Ia-only (118/1493), 0.16 CC-only (24/503); Poisson+Monte Carlo rel. err 0.09–0.21. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 118 SN Ia (subtype audit pending TNS). Volume slice  $z \leq 0.06$ : 1270 SNe total, 69 void (46 Ia) against 1088 wall (645 Ia); apparent slice ratios 0.23 all, 0.26 Ia (no efficiency correction). Threshold grid (SNe): 0.6:56, 0.7:88, 0.8:142, 0.9:220, 1.0:329, 1.1:531. Redshift stability  $z_{\min}$  0.01:142, 0.02:135, 0.025:130, 0.03:126 void. Void median  $z$  0.064 versus wall 0.056 confirms Malmquist gradient. First blinded DR2 Hubble test (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts,  $N = 396$  with 31 void, finite wall  $r/R_{\max} > 1.0$ ; absolute scale sealed; errors include Ia intrinsic scatter 0.12 mag): raw void–wall  $\Delta_{\text{HR}} = +0.057 \pm 0.026$  (stat), mass+ $z$  matched  $+0.023 \pm 0.034$  (27 voids), joint mass slope  $-0.065 \pm 0.011$  (linear) with residual void offset  $+0.037 \pm 0.036$  (mass-step branch  $+0.031 \pm 0.037$  agrees; blinded differences; PV/BBC/nulls pending, no conclusion drawn). FRB error budget (Cat1, following Macquart et al. (2020) and void-decrement simulations (Walker et al. 2024)): per-sightline  $\sigma_{\text{tot}} \sim 150\text{--}200 \text{ pc cm}^{-3}$  (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median  $-3.3$ , std 37.5, p90

68.1, halo 20). Void signal  $\sim 10\text{--}30$ . A  $\delta\text{DM} < 50$  limit at 95 per cent needs  $N_{\text{void}} \gtrsim 36$  ( $\sim 200$  localised total); unlocalised  $z < 0.114$  needs  $\sim 900$  void sightlines, have  $\sim 30\text{--}50$ .

## 5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 2.5%, subdominant). Blinded DR2 Hubble analysis (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, Tripp  $\alpha = 0.14$ ,  $\beta = 3.1$ , blind offset sealed; hand-fit IRSA pilot retired: 5/40 pass QC with  $\chi^2/\text{ndof} \sim 50$  structural mismatch, confirming DR2 adoption):  $N = 396$  with 31 void. Nulls pass: RA-shuffle mean  $+0.004$  sd 0.033 ( $p = 0.48$ ), block bootstrap sd 0.033 vs analytic 0.034, injections 0.00/0.05 recover exactly; mass-step branch agrees with linear ( $+0.031 \pm 0.037$  vs  $+0.037 \pm 0.036$ );  $z_{\min}$  grid stable; selection bound: HR( $z$ ) slope  $\times$  void-wall  $\Delta z$  gives Malmquist bias  $\sim 0.01$  mag, mB-reweighting shifts matched  $\Delta$  by 0.003. Remaining: TNS subtype purge. No unblinding until budget closed. Peculiar velocities bounded: CMB-frame matched  $\Delta_{\text{HR}} = +0.018 \pm 0.034$  vs heliocentric  $+0.023 \pm 0.034$  (shift 0.005);  $z_{\min}$  grid spread  $< 0.002$  mag; diagonal 0.055 mag retained. Systematics budget (Table 3):

**Table 2.** Masked sample composition. Wall means finite  $1.0 < r/R_{\max} < \infty$ ; Outside is  $r/R_{\max} = \infty$  (no void within  $1.2 \max R_{\max}$ ). Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit (91T-like etc.) pending TNS.

| Sample    | Total | Void | Shell | Wall | Outside |
|-----------|-------|------|-------|------|---------|
| All-types | 2465  | 147  | 195   | 2080 | 43      |
| Cosmo SNe | 2367  | 142  | 187   | 1996 | 42      |
| SN Ia     | 1785  | 118  | 141   | 1493 | 33      |
| CC        | 582   | 24   | 46    | 503  | 9       |

**Table 3.** Blinded  $\Delta_{\text{HR}}$  systematics budget (matched analysis). Stat error 0.034 mag dominates; systematics summed below are bounds, not detections. Sources: logs/dr2\_nulls.log, logs/dr2\_pv.log, logs/step\_zcut.log, logs/mask\_buffer\_h.log.

| Source                      | Bound (mag)               | Status  |
|-----------------------------|---------------------------|---------|
| RA-shuffle null             | mean +0.004, $p = 0.48$   | pass    |
| Block bootstrap vs analytic | 0.033 vs 0.034            | agrees  |
| Injection 0.00/0.05         | recovers exactly          | pass    |
| Mass linear vs step         | $ 0.031 - 0.037  = 0.006$ | bounded |
| $z_{\min}$ grid spread      | $< 0.002$                 | bounded |
| CMB-frame shift             | 0.005                     | bounded |
| Mask edge erosion           | 2.5% on fraction          | subdom. |
| PV flow model               | diagonal only             | open    |
| BTS efficiency/BBC          | $\sim 0.01$ mag bound     | bounded |

## 6 DISCUSSION AND CONCLUSIONS

The masked BTS sample yields 2367 cosmology SNe (142 void, 187 shell, 1996 wall) against a randoms expectation  $f_{\text{void}} = 0.190$ : an apparent void/wall ratio 0.24 overall (0.26 Ia, 0.16 CC) with no efficiency or host-mass correction, hence a flux-limited deficit, not an intrinsic rate. The in-mask transient fraction  $2465/6873 = 0.359$  exceeds the geometric sky fraction 0.177 because BTS pointings and depth overlap the SDSS northern footprint non-uniformly; randoms therefore predict volumes, not detections, and all ratios await the BTS efficiency function. The void redshift offset (median 0.064 vs 0.056) is the Malmquist gradient in the same data. First blinded DR2 test ( $N = 396$ , 31 void, finite wall): raw  $\Delta_{\text{HR}} = +0.057 \pm 0.026$  falls to  $+0.023 \pm 0.034$  mass+ $z$  matched (27 voids); joint fit recovers the mass slope ( $-0.065 \pm 0.011$ ) with residual void offset  $+0.037 \pm 0.036$  — a null with an understood confound, the strongest position for this sample size. Limitations: 1-deg mask quantization, pending TNS subtype purge and spec- $z$  audit, pending peculiar-velocity flow-model and host-mass-step refinements, and an FRB arm that is a forecast only (Catalog 2 required). Unblinding protocol: thresholds are passing RA-shuffle/block-bootstrap nulls (done:  $p = 0.48$ , bootstrap validates error), a closed systematics budget, and frozen cuts; the blind key rotates at unblinding; until then no Hubble-offset conclusion is drawn.

## ACKNOWLEDGEMENTS

H. D. is a Higher Secondary Certificate student. This work used public archives only and received no institutional funding. The author thanks the VAST, ZTF, CHIME/FRB, SDSS, and NASA-Sloan Atlas teams for making their data and software public. AI agents (Muse Spark 1.3) assisted in building analysis codes, LaTeX files, and figures; however, the scientific analysis, interpretations, and conclusions remain the sole responsibility of the author, who independently verified all generated numbers by re-execution.

## 7 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-29 UTC, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3\_match\_v2.parquet (sha256 1a162c74...af77530) with scripts and randoms (seed 42/43/44) released with a Zenodo DOI on acceptance; repository public on acceptance.

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